Excitation Dynamics with Quantum Transitions in Neutral Argon Clusters

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1. Introduction

We use Excitation Dynamics with Quantum Transition in whole Diabatised - Diatomic In Molecules (Di-DIM) basis to study excitation evolution in small neutral argon cluster, ($N = 3, 7$), which has not been done previously. We further investigate the fragmentation channels associated to the relaxation following excitation in the lowest singlet excited state of the 4s band. We show the effect of highly degenerate states and their effect on relaxation dynamics. We observe relaxation to a lower triplet state in Ar_3^* and Ar_7^* clusters. We observe emission of excited atom to be the dominant dissociation channel with the formation of a ground state cluster. This is followed by formation of an excimer (Ar_2^*) and atomisation. The observation of excimer with a bigger excited cluster is dependent on the initial cluster size. We do not observe formation of a stable excited trimer isomer using Di-DIM. We observe that the nature and the number of states used as the input for DIM, influences significantly the outcome of the relaxation dynamics.

2. Simulation Details

2.1. Diabatisation Scheme

In our previous work Dhiman *et al.*[1] we show that the potentials used by Naumkin *et al.*[2] in standard DIM - where dimer PECs are calculated using MRCI - are not sufficient to describe the lowest energy excited trimer. The Ar_3^* equilibrium geometry has C_{2v} symmetry rather than $D_{\infty h}$ symmetry, as expected from the more elaborate Hole-Particle Pseudopotential (HPP) method given by Duplaa *et al.*[3, 4]. The reasoning for this difference is potentially due to the strong coupling from the higher excited states (see my dissertation [10]), which have been noted previously. In this approach we perform dibatisation using an *ad-hoc* scheme based on the introduction of two pseudo-excited

states. We validate the diabatisation with trimer results obtained using HPP method. The effect of diabatisation is illustrated in the figure 1.

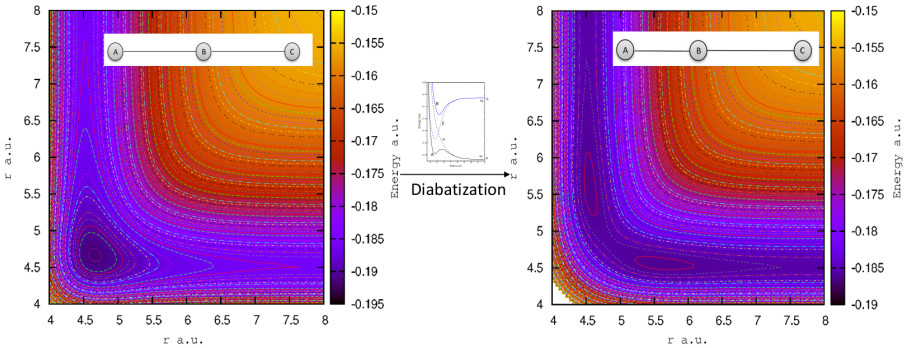


Figure 1: Diabatisation effect on the lowest Ar_3^* potential energy surface. The left figure shows the trimer PES obtained using DIM to be in C_{2v} symmetry and the right figure shows the breaking of symmetry with diabatisation leading to formation of trimer in $D_{\infty h}$ symmetry.

2.2. Non-Adiabatic Dynamics with Quantum Trajectories

We present a theoretical model to study the time evolution of the Rydberg states of excited neutral rare-gas clusters. Our model is based on previous work on ionic clusters [5, 6, 7]. The Potential Energy surface (PES) obtained *on-th-fly* using Di-DIM method. The evolution of the electronic wave packet is based on the method proposed by Janecek *et al* [8] using the whole Di-DIM basis. The unitary time-displacement operator is evaluated using Magnus expansion [9] in first order expansion, where we consider the perturbation to be very small for a small time-step. Here, we give an algorithm for coefficient evolution in the basis of time-averaged Hamiltonian to minimise the perturbations. The SOC is included within Di-DIM method. The Fewest Switching Surface Hopping (FSSH) is employed for hops between adiabatic surfaces. The transition between the adiabatic states is done using surface-hopping algorithm proposed by Tully. Finally, the propagation of nuclei is done classically using Beeman method on adiabatic PES. This method is validated by analytically implementing it for a two state system, see the dissertation for more details [10].

3. Key Results

The dynamics is performed to observe collision and dissociation of Neutral Argon system. In collision process we observe the excitation transfer and decay based on the collision energy. In dissociation dynamics, we observe emission of excited Argon species along neutral species with relaxing ground state system, not observed previously. As an example, figure 2 shows an excited Ar_7^* system leading for formation of a ground state Argon dimer with emission of an excited atom in $J = 2$ state of the ^{3}P(4s) configuration.

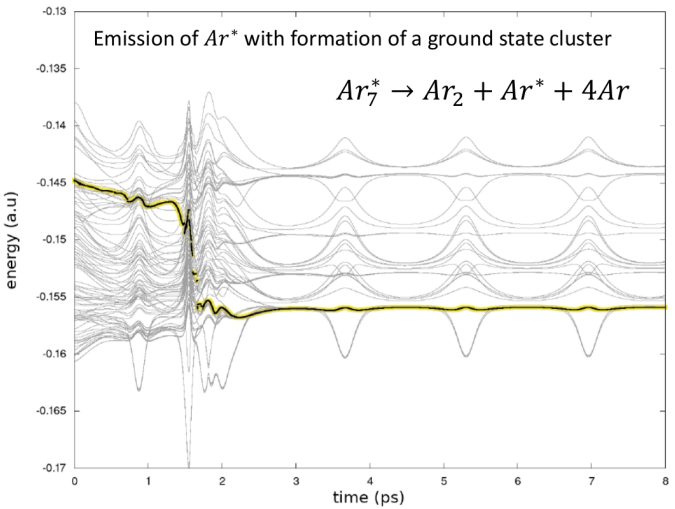


Figure 2: A trajectory on the PEC leading to formation of a slowly oscillating ground state dimer. This dynamics is done for 10 ps and we show first 8 ps. Ad hoc states are not shown here which are higher in energy. The highlights shows the 4*s dissociation limit of the excited atom and the corresponding, almost flat, PECs. Final state is* $J = 2$

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